



RCA & FIX

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

Fleet Offline Depot Metadata Mismatch

Broadcom's offline depot metadata zip references a superseded 9.0.2.0 patch binary that no longer exists on the downloads portal. Fleet downloads the 9.0.2.0100 replacement but fails on apply due to an internal path mismatch.

LCMPATCHUPDATE16002

OFFLINE DEPOT

FLEET MANAGEMENT

9.0.2.0100

METADATA MISMATCH

VCF 9.0

VMware Cloud Foundation

Fleet Management Offline Depot Metadata Mismatch — 9.0.2.0 vs 9.0.2.0100

Prepared by: Virtual Control LLC **Date:** May 26, 2026 **Undocumented Issues Reference:** Issue #47

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1. Problem Statement

The VCF offline depot metadata zip (`vcf-9.0.2.0-offline-depot-metadata.zip`) downloaded from the Broadcom Support Portal contains a `productVersionCatalog.json` that references the Fleet Management patch binary `VCF-OPS-Lifecycle-Manager-9.0.2.0.25137839.patch` . Broadcom superseded that binary with `VCF-OPS-Lifecycle-Manager-9.0.2.0100.25362033.patch` (released April 27, 2026) and removed the 9.0.2.0 binary from the downloads portal. The metadata zip was never updated to match.

This prevents Fleet Management from being upgraded from 9.0.1 to 9.0.2.0100 via the standard UI patch workflow when using an offline depot. Without Fleet at 9.0.2.0100, VCF Automation 9.0.2 cannot be deployed because Fleet's policy layer caps deployable product versions to its own version.

2. Environment

Component	Detail
Fleet Management	9.0.1.0, build 24960371
Target version	9.0.2.0100, build 25362033
Depot type	URL-based offline (<code>https://192.168.1.160:8443</code>)
Metadata zip	<code>vcf-9.0.2.0-offline-depot-metadata.zip</code> (188,858 bytes, sequence 33)
Patch binary on portal	<code>VCF-OPS-Lifecycle-Manager-9.0.2.0100.25362033.patch</code> (1.53 GB)

Component	Detail
Patch binary in catalog	VCF-OPS-Lifecycle-Manager-9.0.2.0.25137839.patch (no longer available)

3. Symptoms

After downloading the Fleet Management 9.0.2 patch via Binary Management and clicking Install Patch, the operation fails immediately:

```
Error Code: LCMPATCHUPDATE16002
Failed to apply VCF Operations Fleet Management patch.
Error while creating local patch file
```

In `/var/log/vrlcm/vmware_vrlcm.log`:

```
LCM patch uRL: https://lcm.lab.local/repo/productPatchRepo/patches/vrlcm/9.0/
VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch
ContentDownloadController -- Error occurred when processing the download content.
PatchUpdateTask -- Error occurred in downloading patch from content repo
java.lang.NullPointerException: null
    at PatchUpdateTask.createLocalFile(PatchUpdateTask.java:199)
```

The URL Fleet requests (`...9.0.2.0100.patch`) does not match the URL stored in its internal content repository database (`...9.0.2.0.patch`).

4. Root Cause Analysis

Root cause: Broadcom updated the Fleet Management patch binary from version 9.0.2.0 (build 25137839) to 9.0.2.0100 (build 25362033) on their downloads portal but did not update the offline depot metadata zip. The zip filename, file size, and `sequenceNumber` are all unchanged, giving no indication that it is stale.

The version mismatch creates a three-way conflict inside Fleet:

Component	Version referenced	Source
<code>productVersionCatalog.json</code> (depot metadata)	<code>9.0.2.0.25137839</code>	Broadcom metadata zip
Patch binary (actual file downloaded)	<code>9.0.2.0100.25362033</code>	Broadcom downloads portal
<code>patchProductRegistry.json</code> (extracted from .patch)	<code>9.0.2.0100</code>	Internal manifest inside the .patch file

What happens during download:

1. Fleet reads the catalog, sees `VCF-OPS-Lifecycle-Manager-9.0.2.0.25137839.patch`, and fetches it from the depot (the operator must rename the actual 9.0.2.0100 file to match the catalog filename).
2. Fleet validates the download checksum against the patch's **internal** manifest (not the catalog checksum) and it passes.

3. Fleet extracts the `patchProductRegistry.json` from inside the `.patch` file and stores it in the CMS database. This registry correctly identifies the patch as version `9.0.2.0100`.
4. Fleet stores the extracted patch payload (1.2 GB ZIP) in the content repository under blob UUID `7a2bdca3-....`.
5. Fleet creates a URL mapping in `vm_cr_contentdownloadurl` using the **catalog-derived** version: `/productPatchRepo/patches/vr1cm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0.patch`.

What happens during patch apply:

6. Fleet reads the `patchProductRegistry.json`, which says the payload file is `VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch`.
7. Fleet asks the content controller for URL `/productPatchRepo/patches/vr1cm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch`.
8. The content controller looks up this URL in `vm_cr_contentdownloadurl` and finds **no match** (the stored URL has `9.0.2.0` not `9.0.2.0100`).
9. `NullPointerException` at `PatchUpdateTask.createLocalFile()` line 199.

5. Resolution

Fix: Update one row in `vm_cr_contentdownloadurl` on the Fleet appliance to change the URL path from `9.0.2.0` to `9.0.2.0100`. This aligns the content controller's URL mapping with what the `patchProductRegistry.json` expects.

Prerequisites

- SSH access to Fleet appliance as root
- Fleet snapshot taken before attempting patch (standard pre-patch requirement)

Procedure

```
-- SSH to Fleet appliance, then:
cd /tmp && sudo -u postgres /opt/vmware/vpostgres/current/bin/psql -d vr1cm

-- Step 1: Verify current (broken) state
SELECT vmid, url, content FROM vm_cr_contentdownloadurl
WHERE url LIKE '%Lifecycle-Manager-Appliance-9.0.2%';
-- Expected: .../VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0.patch

-- Step 2: Fix the URL
UPDATE vm_cr_contentdownloadurl
SET url = '/productPatchRepo/patches/vr1cm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch'
WHERE url = '/productPatchRepo/patches/vr1cm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0.patch';
-- Expected: UPDATE 1

-- Step 3: Verify
SELECT vmid, url, content FROM vm_cr_contentdownloadurl
WHERE url LIKE '%Lifecycle-Manager-Appliance-9.0.2%';
-- Expected: .../VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch
```

```
# PowerShell on workstation (via Invoke-VMScript):
$fleet = Get-VM 'fleet'
Invoke-VMScript -VM $fleet -GuestUser 'root' -GuestPassword '<password>' -ScriptType Bash -ScriptText '@'
```

```
cd /tmp && sudo -u postgres /opt/vmware/vpostgres/current/bin/psql -d vrlcm -c "  
UPDATE vm_cr_contentdownloadurl  
SET url = '/productPatchRepo/patches/vrlcm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch'  
WHERE url = '/productPatchRepo/patches/vrlcm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0.patch';"  
'@
```

After the update, retry the patch: **Settings > System administration > System patches > Install patch > fleet management > Install**.

6. Verification

After running the UPDATE, verify the content controller can resolve the corrected URL:

```
# On Fleet appliance:  
curl -sk -o /dev/null -w '%{http_code}' \  
  "https://localhost/lcm/repo/productPatchRepo/patches/vrlcm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-  
  9.0.2.0100.patch"  
# Expected: 200
```

After successful patch apply:

```
# Confirm Fleet version is now 9.0.2.0100:  
curl -sk "https://localhost/lcm/bootstrap/api/status" | python3 -m json.tool | grep version  
# Expected: "version": "9.0.2.0100"
```

7. Rollback

If the patch apply still fails after the URL fix:

```
-- Revert the URL change:  
UPDATE vm_cr_contentdownloadurl  
SET url = '/productPatchRepo/patches/vrlcm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0.patch'  
WHERE url = '/productPatchRepo/patches/vrlcm/9.0/VCF-OPS-Lifecycle-Manager-Appliance-9.0.2.0100.patch';
```

A pre-patch snapshot of Fleet is available for full revert if needed.

8. Document Information

Field	Value
Document Title	Fleet Management Offline Depot Metadata Mismatch
Author	Virtual Control LLC
Date Issued	May 26, 2026
Environment	VCF 9.0.1 / Fleet Management 9.0.1.0 build 24960371
Related Issues	Undocumented Issues Reference Issue #47

Field	Value
Broadcom KB searched	LCMPATCHUPDATE16002 , offline depot metadata 9.0.2.0100 , Fleet patch validation failed createLocalFile , productVersionCatalog 9.0.2 stale — no matching KB as of 2026-05-26

Revision History

Version	Date	Change
1.0	May 26, 2026	Initial issue documentation. Root cause identified and fix verified.

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